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Studierichting: Informatica
Oefeningenreeks 5G nr. 9

$$y^2 = x^4 + x^5$$

$$\Rightarrow y = \pm\sqrt{x^4 + x^5}$$

$$V = \pi \int_{-1}^0 \left(\sqrt{x^4 + x^5} \right)^2 dx = \pi \int_{-1}^0 (x^4 + x^5) dx$$

$$= \pi \left[\frac{x^5}{5} + \frac{x^6}{6} \right]_{-1}^0$$

$$= \pi \cdot \left(\frac{1}{5} - \frac{1}{6} \right)$$

$$= \pi \cdot \left(\frac{6}{30} - \frac{5}{30} \right) = \frac{\pi}{30}$$

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References